

Question 8 – Stage 1

A cube and a sphere have the same volume and same centre.

If $p\%$ of the surface area of the cube lies outside the sphere, what is p (to 3 sig.figs.)?

Working

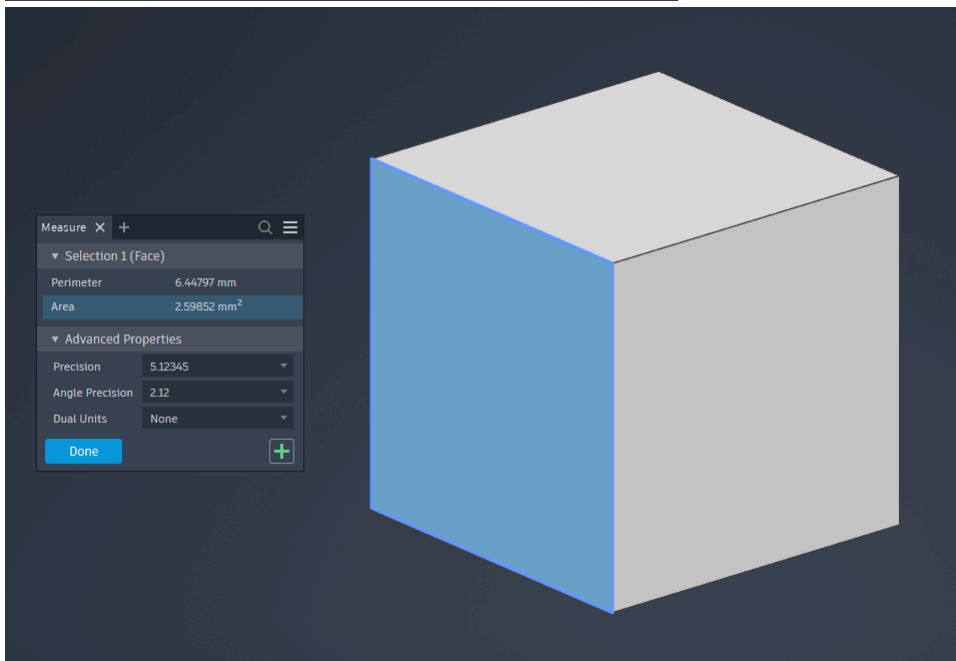
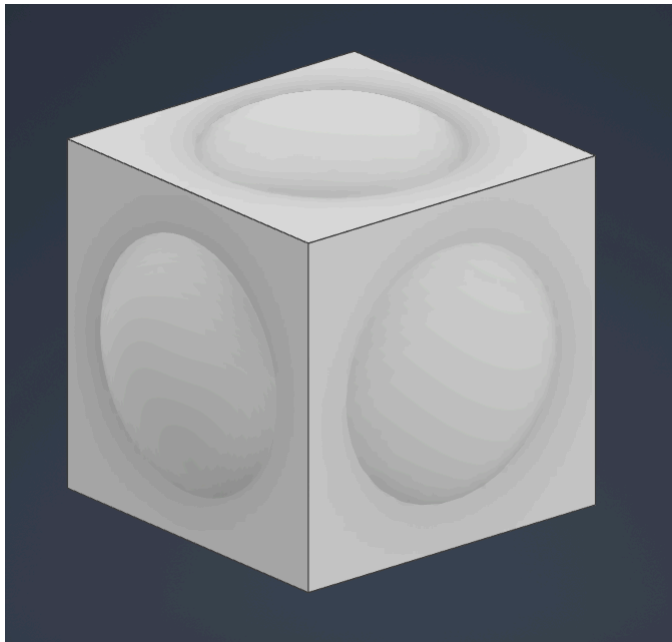
Find Side Length for unit sphere

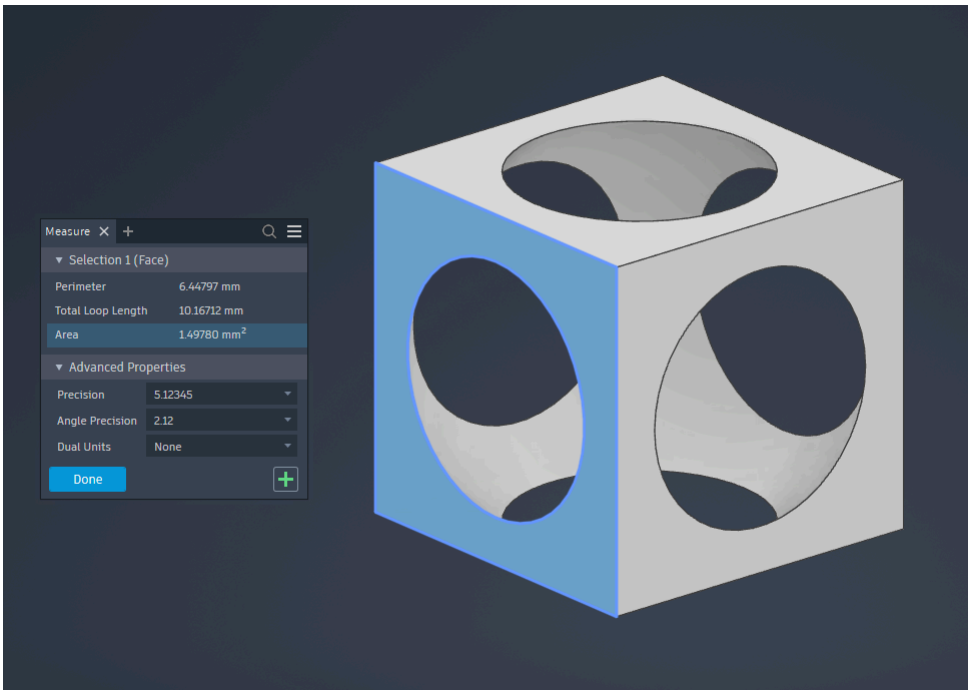
x = cube side length r = sphere radius

$$\begin{aligned}x^3 &= \frac{4}{3}\pi r^3 \\x &= \left(\frac{4}{3}\pi r^3\right)^{\frac{1}{3}} \\ \text{let } r &= 1 \\x &= \left(\frac{4}{3}\pi\right)^{\frac{1}{3}} \\x &\approx 1.611991954\dots\end{aligned}$$

Model in Autodesk Inventor to measure the area

[Inventor File](#)





Calculate Percentage

$$\begin{aligned}
 &A \text{ is area of one face} \\
 &A \approx 2.59852 \\
 &A_{Cut} \approx 1.49780 \\
 &\frac{A_{Cut}}{A} \approx 0.576405 \\
 &\approx 57.6\%
 \end{aligned}$$